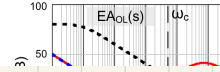
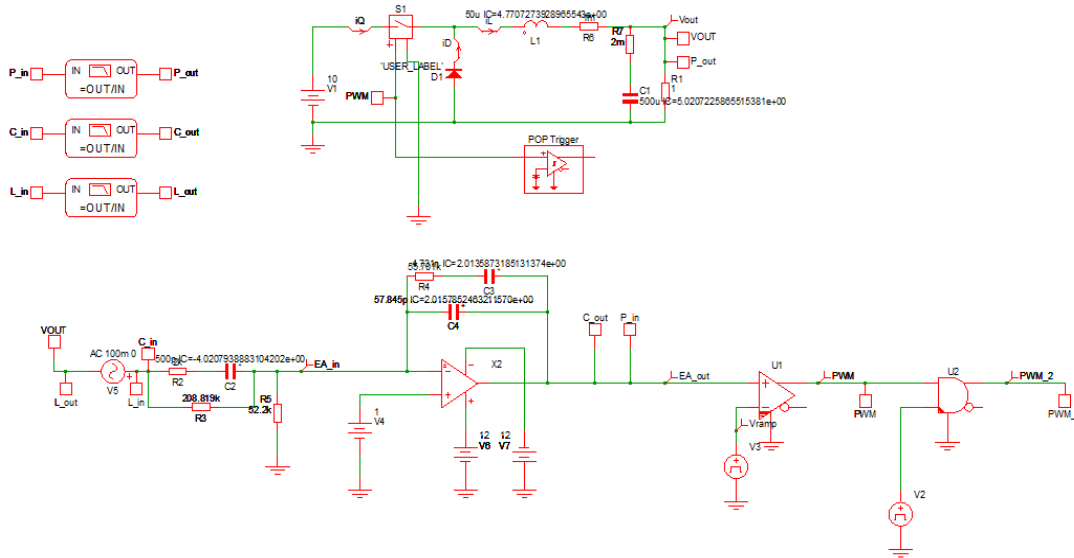




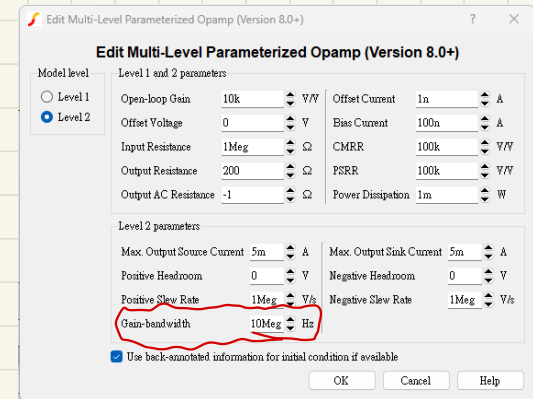
(1) (15%) Define the spec. of amplifier including dc gain and 3dB bandwidth first, which achieves designed compensation transfer function and loop phase margin in HW3. Then use simulation tool to verify. Hint: Use level 2 model of amplifier to simulate compensation transfer function with external compensation components and loop gain. Please also show the test circuit for verification.

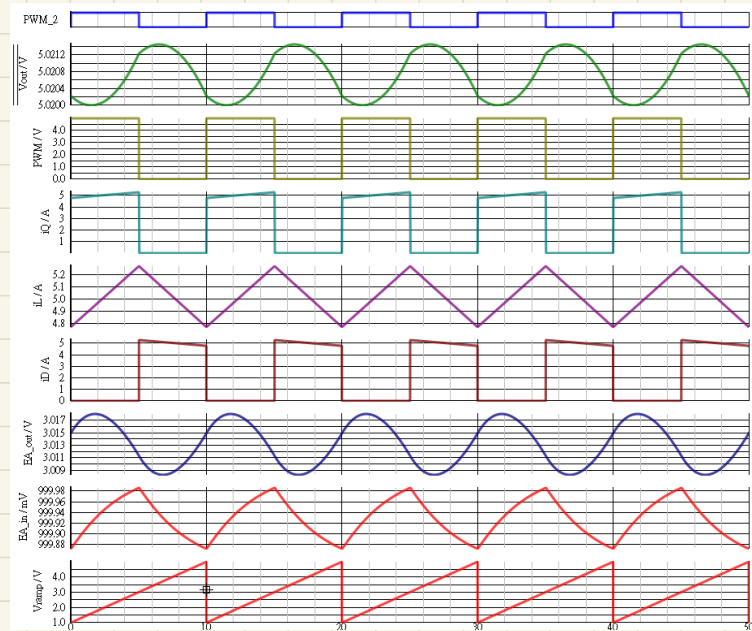
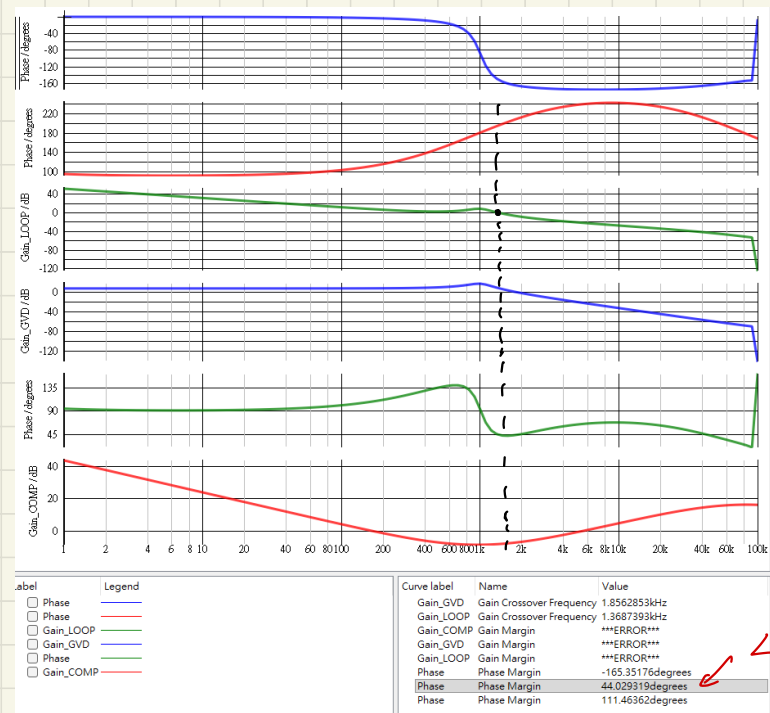


Test Circuit (same as HW3)



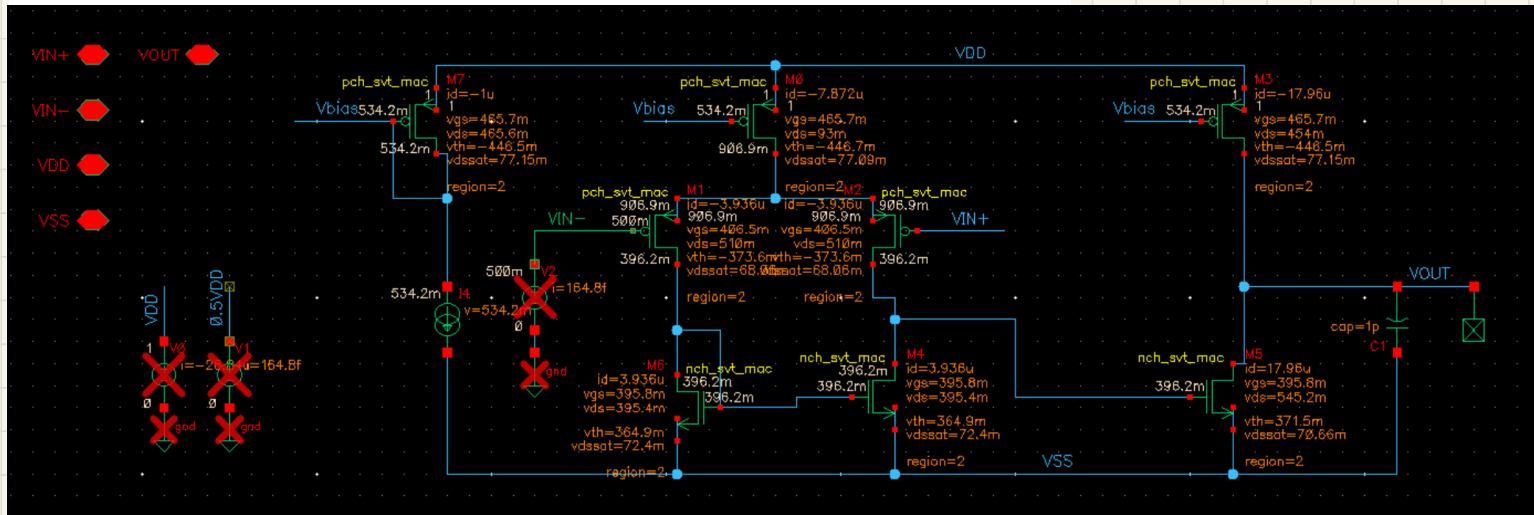
spec for amplifier





40° phase margin

(2) (35%) Please design the transistor level of the amplifier used for compensator. Use the provided FinFET process. Please write down the design procedure. Then use simulation tool to verify that your design meets the spec. of amplifier in (1).

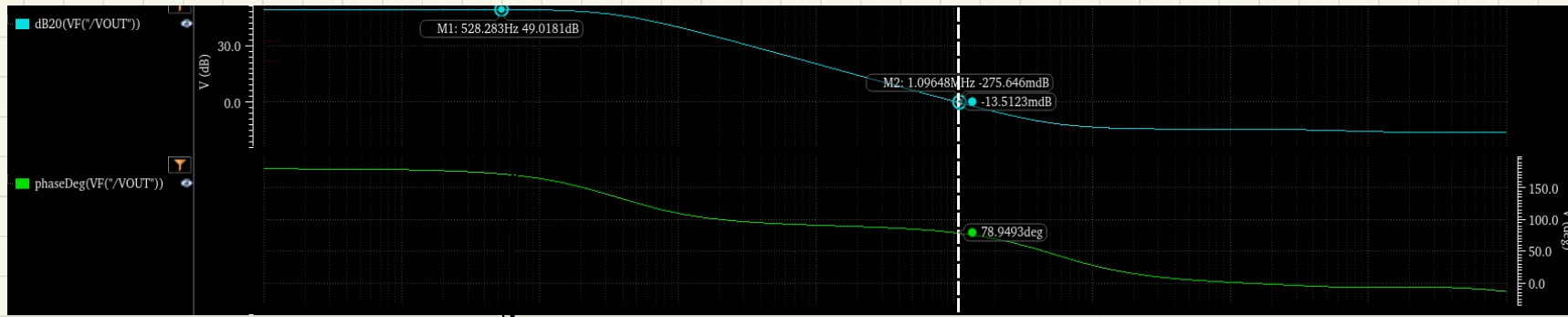


Design Procedure : ① Set current mirror ratio of two stage
($I_1 = I_2 = 18$)

② Check current mirror transistor M1, M_6 is operating saturation region
(increase length)

③ check output stage transistor M_3, M_5 in saturation region
(increase width of M_5)

④ Add miller capacitor to reduce unity gain bandwidth

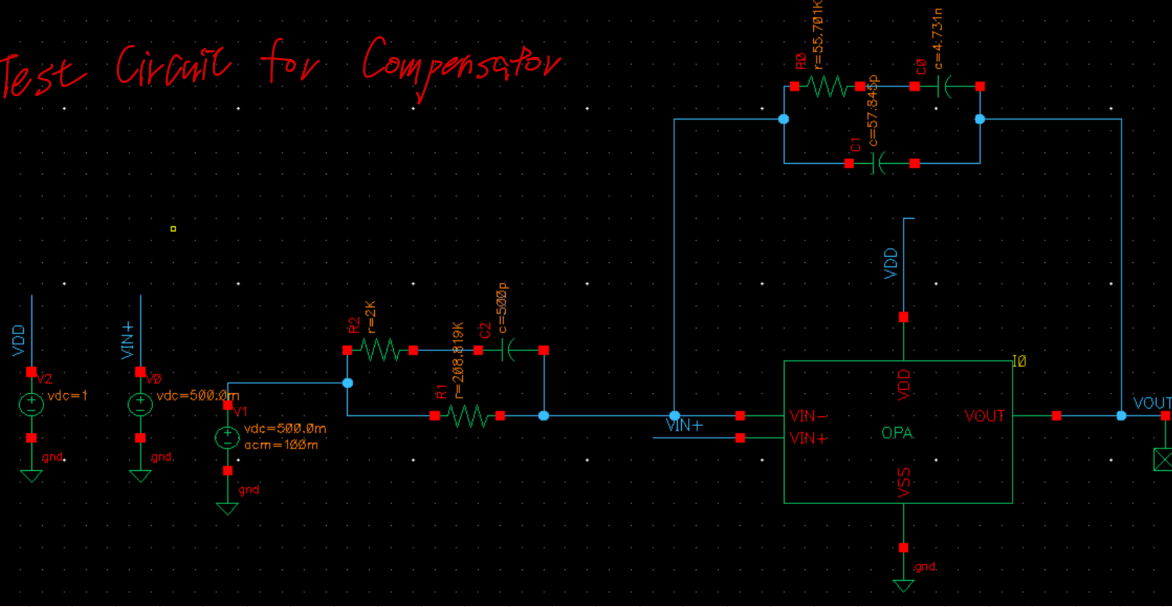


↓ open-loop gain = 281.8

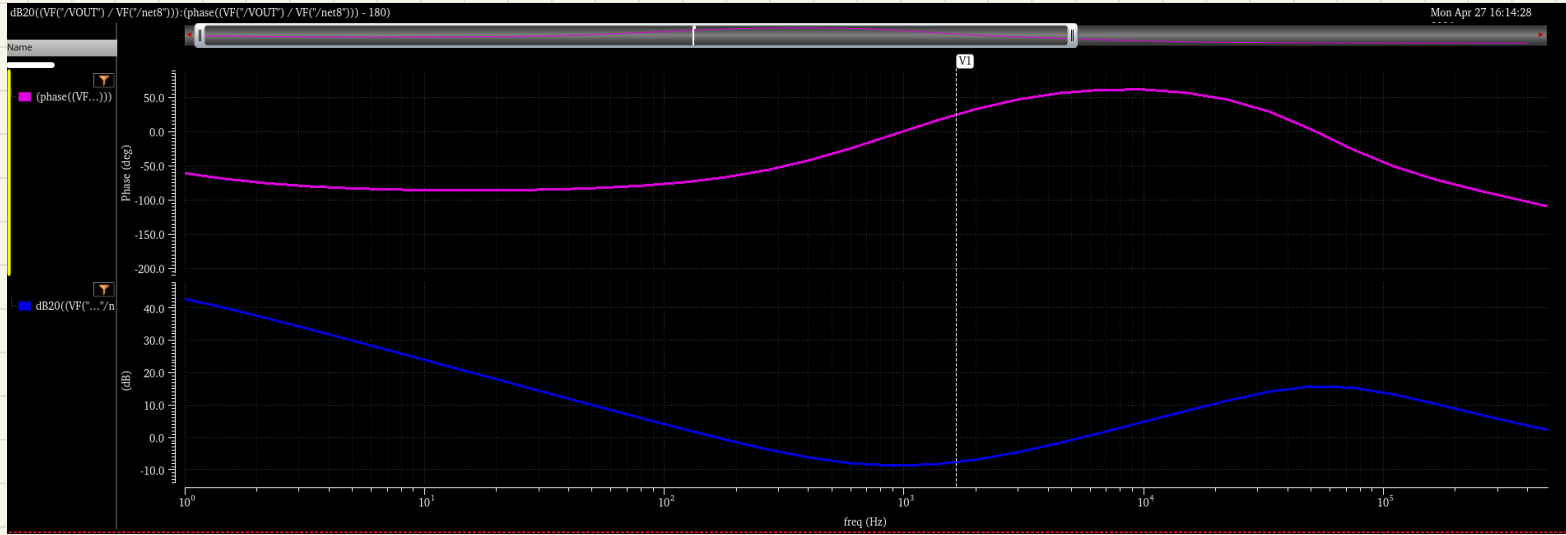
unity gain-bandwidth 1.09 MHz

PM = 78.9°

Test Circuit for Compensator



Compensator AC analysis



frequency below 2kHz, the comp AC analysis is almost the same

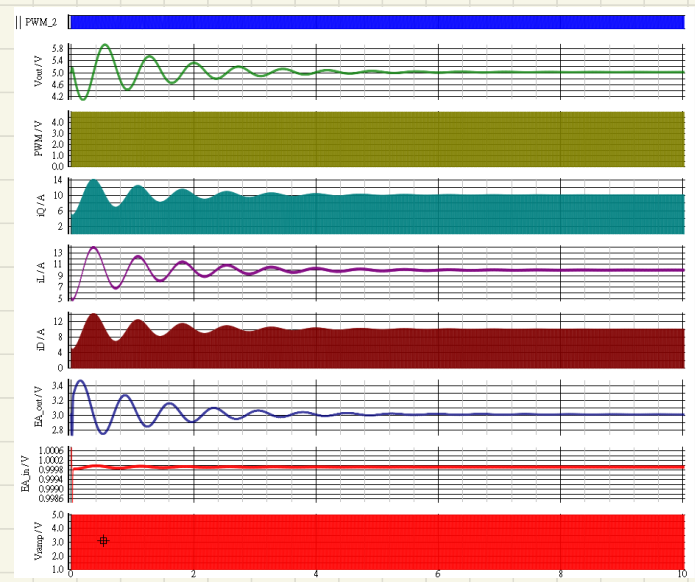
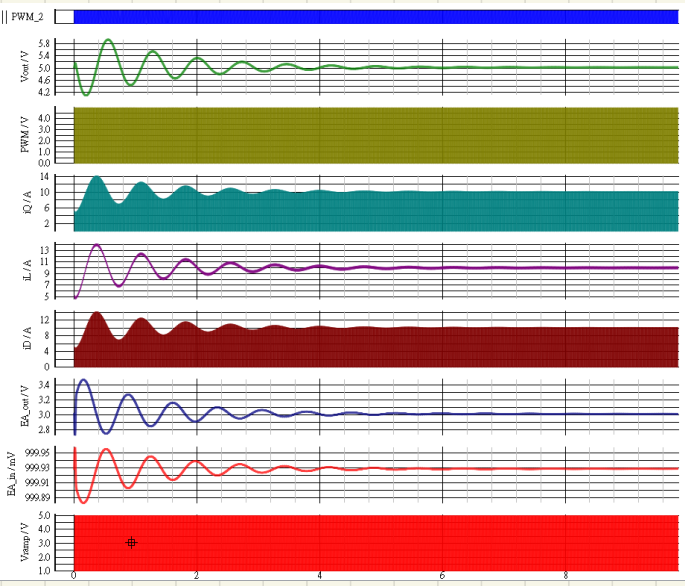
■ You get extra points if you compare the simulated load transient responses, loop gains, and compensator gains of designed Voltage-Mode Controlled Buck Converter using level 1, 2, 3 models of amplifier.

(6% for each comparison item. Multiply by 1 (2) if you compare level 1, 2, (3))

level 1: behavior model (ideal). level 2: behavior model (finite

level 1 (5 A → 10 A)

level 2



level $\geq$

